

- if
- else
- if else
- nested if
- if elif else

```
In [1]: # Check if the number if even or not
x=4
r=x%2
if r==0:
    print('Even Number')
```

Even Number

```
In [2]: x=4
r=x%2
if r==0:
    print('Even Number')
```

```
Cell In[2], line 3
    if r==0;
        ^
SyntaxError: invalid syntax
```

```
In [3]: #indentation can be atleast one space for print statement
x=4
r=x%2
if r==0:
    print('Even Number')
```

Even Number

```
In [4]: # if the condition not match then no output will be displayed
x=5
r=x%2
if r == 0:
    print('Even Number')
```

```
In [5]: #condition is not met however the code prints the last statement
x = 5
r=x%2
if r==0:
    print('Even Number')
print('Odd Number')
```

Odd Number

```
In [6]: x=8
r=x%2
if r == 0:
    print('Even Number')
print('Odd Number')
```

Even Number
Odd Number

```
In [7]: #Two if conditions will be executed no matter with the conditional value
x= 9
r=x%2
if r == 0:
    print('Even Number')
if r != 0:
    print('Odd Number')
```

Odd Number

```
In [8]: # if else condition
x= 9
r= x%2
if r==0:
    print('Even Number')
else:
    print('Odd Number')
```

Odd Number

```
In [9]: x=8
r=x%2
if r==0:
    print('Even Number')
else:
    print('Odd Number')
```

Even Number

```
In [10]: #Nested if
x= 3
r=x%2
if r==0:
    print('Even number')

    if x>5:
        print('Greater than number 5')
else:
    print('Odd Number')
```

Odd Number

```
In [11]: x = 4
r = x%2
if r == 0:
    print('Even Number')

    if x>5:
        print('Greater than number 5')
else:
    print('Odd Number')
```

Even Number

```
In [12]: x= 4
r=x%2
if r==0:
    print('Even Number')
    if x>5:
        print('Greater than number 5')
else:
```

```
        print('Smaller than number 5')  
    else:  
        print('Odd Number')
```

Even Number
Smaller than number 5

```
In [13]: x = 1  
if x==1:  
    print('one')  
if x==2:  
    print('Two')  
if x==3:  
    print('Three')  
if x==4:  
    print('Four')
```

one

```
In [14]: x=1  
if x==1:  
    print('One')  
elif x==2:  
    print('Two')  
elif x==3:  
    print('Three')  
elif x==4:  
    print('Four')
```

One

```
In [16]: x=10  
if x==1:  
    print('One')  
elif x==2:  
    print('Two')  
elif x==3:  
    print('Three')  
elif x==4:  
    print('Four')  
  
else:  
    print('Number not found')
```

Number not found

Loops

```
In [17]: i = 1  
while i<=5:  
    print('Data Science')  
    i= i+1
```

Data Science
Data Science
Data Science
Data Science
Data Science

```
In [19]: i = 1
while i<=5:
    print('data science',i)
    i= i+1
```

```
data science 1
data science 2
data science 3
data science 4
data science 5
```

```
In [20]: i = 5
while i>=1:
    print('data science')
    i = i-1
```

```
data science
data science
data science
data science
data science
```

```
In [21]: i = 5
while i>=1:
    print('data science',i)
    i= i-1
```

```
data science 5
data science 4
data science 3
data science 2
data science 1
```

```
In [23]: i = 1
while i<=5:
    print('data science')
    j=1
    while j<=4:
        print('technology')
        j=j+1
    i=i+1
    print()
```

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

data science
technology
technology
technology
technology

In []: